

One control core. Power conversion at every scale.

OpenVVF separates a reusable dual-MCU control and safety core from the power hardware. The same interfaces support smaller user-supplied stages, the 465 A-class Size 2 inverter, a released 15 kW DC/DC module, and a future 1,400 A chassis. Hardware, firmware, safety analysis, and test evidence are published openly.



The Size 2 control and power stack is built, bench-tested, and undergoing validation.

200-450 V

Size 2 DC-bus class by capacitor selection

465 A

Continuous design rating · 600 A peak (60 s)

10

Published test reports, raw telemetry linked

PMSM + induction

Instrument-verified motor self-commissioning

■ Modular by design

The controller is largely **power-stage agnostic**. PWM and protection interfaces can connect to lower-power or user-supplied gate-drive stages with appropriate sensing and gate-drive scaling, while the same core extends to higher-voltage and higher-current hardware.

■ Built and bench-tested

Size 2 has completed a **180 V bring-up**, a **one-hour endurance run**, a **20-minute reversal stress test**, and instrument-verified **motor self-commissioning**. Its dual-MCU architecture provides **six safe-state pathways**.

■ Open research at UCSC

Professor **Keith Corzine's Smart Power Lab** at UC Santa Cruz develops the project in the open. Schematics, firmware, safety analyses, assembly data, and **interactive telemetry logs** are public at docs.openvvvf.org.

MODULAR POWER BLOCKS

Scale down or up around one control core

CONTROL MODULE · Dual-MCU · FOC motor control + independent safety coprocessor · USB-B, dual CAN FD, and expansion I/O · RTE Studio + Telemetry Viewer



Lower-power stages

Interface-ready for smaller or user-supplied gate-drive hardware

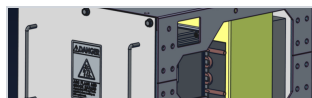
■ **SCALABLE INTERFACE**



Size 2 inverter

200–450 V class · 465 A continuous design rating

■ **IMPLEMENTED, UNDER TEST**



Size 2 DC/DC

15 kW · 450 V class · reuses Size 2 boards and power stage

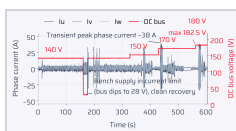
■ **RELEASED DESIGN**



Size 3

Up to 1,200 V · 1,400 A · modular liquid-cooled enclosure

■ **IN DEVELOPMENT**



Live telemetry, not promises

Test reports include interactive logs for the browser-based Telemetry Viewer. Scan to open the 180 V bring-up: 140 to 180 V bus steps, approximately 38 A peak current, and clean recovery from a supply current-limit event.



SUPPORT THE SMART POWER LAB

Support open power electronics at UC Santa Cruz.

UCSC can structure support as a research gift, sponsored research, or an in-kind contribution. Technical contact: Professor Keith Corzine, corzine@ucsc.edu.

CURRENT SUPPORTERS

